Pratham Modi

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Machine Learning Engineer | Agentic AI, Python, FastAPI, RAG, PyTorch, LangGraph

Summary

Machine Learning Engineer with 2+ years of production experience designing and deploying agentic AI architectures, RAG pipelines, and high-throughput data processing backends. Built autonomous multi-agent orchestration systems (LangGraph, CrewAI) cutting injected tokens by 38%, optimized vector retrieval over 10K+ documents at Playpower Labs, and engineered resilient event streaming with Kafka and Redis at C3iHub (IIT Kanpur).

Technical Skills

AI, ML & Agents: Multi-Agent Systems, LangGraph, CrewAI, Agentic AI, RAG Pipelines, Vector Databases (Qdrant, ChromaDB), Prompt Engineering, LLM Evaluation
Languages & Frameworks: Python, FastAPI, Node.js, SQL, JavaScript, TypeScript
Backend & Data Streaming: Microservices, Kafka, Redis Streams, PostgreSQL, MongoDB, Event-Driven Architecture
Cloud & MLOps: AWS (EC2, S3, EKS), Docker, Kubernetes, CI/CD, GitHub Actions, OpenTelemetry, Prometheus, Grafana

Experience

C3iHub, IIT Kanpur Jul 2025 – Present
Software Development Engineer (Backend & Infrastructure) Bengaluru / Kanpur, India

- Architected core platform backend and microservices using Express.js, MongoDB, Redis, and Socket.IO, serving 50K+ users and maintaining 99.9% uptime under 10,000+ concurrent connections.
- Engineered event-driven notification pipeline handling 75K+ daily events at 99.5% delivery reliability using Kafka, BullMQ, and Redis Pub/Sub with consumer-side deduplication.
- Improved incident response time 85% by deploying distributed tracing across 15+ microservices with OpenTelemetry, Prometheus, and Grafana; defined alerting SLOs across cross-functional teams.
- Led engineering of official IITK platform onboarding 10K+ participants pan-India; productized into multi-tenant SaaS serving public and private enterprise deployments with RBAC.

Playpower Labs Nov 2024 – Jun 2025
Software Development Engineer (Full-Stack & AI Services) Bengaluru, India

- Increased user session depth by 35% by integrating ChromaDB vector search over 10K+ documents via a RAG pipeline, achieving sub-200ms retrieval latency at p95.
- Cut document retrieval latency by 47% (340ms to 180ms) with a PDF ingestion pipeline using 512-token chunking, OpenAI embeddings, and HNSW-indexed ChromaDB.
- Developed AI tutoring platform using Next.js with SSR/ISR and React Server Components alongside FastAPI microservices, supporting 500+ concurrent WebSocket sessions with sub-100ms response times.

Projects

Mnemoniq: Multi-Tier Memory AI Agent | *FastAPI, LangGraph, PostgreSQL, Qdrant, Redis*

- Architected 3-tier memory system (working, episodic, graph) for conversational AI orchestrated via LangGraph, cutting context tokens 38%.
- Combined Qdrant vector retrieval with NetworkX graph traversal behind a FastAPI and SSE backend with PostgreSQL and Redis sessions.

DeployMind: AI-Powered GitOps Platform | *Python, FastAPI, CrewAI, Redis, Docker, Kubernetes*

- Built GitOps platform automating deployments from GitHub to EC2 and Kubernetes (EKS) via a 3-agent AI pipeline using CrewAI with security gates.
- Implemented rolling, canary, and blue-green deployment strategies with Redis-backed queue, distributed locking, and 200+ automated tests.

Education

Pandit Deendayal Energy University 2021 – 2025
Bachelor of Technology in Computer Science and Engineering CGPA: 9.20 / 10.0